

Real-time Multi-modal Emotion Recognition for Indian Languages

[Summer 2024]

Broad Research Goal:

To dive deep into real-time multimodal and multilingual emotion analysis in conversations.

Current State:

- Very few works on real-time emotion analysis. There is extensive research on emotion recognition in conversation (ERC), emotion-cause prediction (ECP), and emotion-cause pair extraction (ECPE, a multi-task scenario of emotion recognition and cause prediction).
- All the SOTA work is on English conversation analysis. We aim to tackle multi-lingual code mixed conversations for emotion recognition.

Motivation (why the idea has potential):

- Real-time emotion analysis systems can assist conversation participants (including generative models) in understanding the emotions of others and generate responses according to the participant's emotional state.
- Exploring emotion recognition in audio-only scenarios as that is what is possible to obtain during real-time inference. Audio can be plagued with noise, so robust models can be designed to perform at par with traditional models, which assume no noise in the input audio.
- Previous research has shown that using audio and text gives the best results in emotion recognition systems. We can extend the same principles to real-time settings as well.

Research Question:

- Can we build a real-time multilingual code-mixed emotion analysis system for India that works on noisy audio?

Initial/Rough Ideas/thoughts to explore:

- Use existing ideas from the below papers and apply them to real-time scenarios:
 - Abhinav Joshi, Ashwani Bhat, Ayush Jain, Atin Singh, and Ashutosh Modi. 2022. COGMEN: COntextualized GNN based Multimodal Emotion recognition. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 4148–4164, Seattle, United States. Association for Computational Linguistics.

- Yuzhao Mao, Guang Liu, Xiaojie Wang, Weiguo Gao, and Xuan Li. DialogueTRM: Exploring Multi-Modal Emotion Dynamics in Conversations.
- Wenxiang Jiao, Michael R. Lyu, and Irwin King. 2019. Real-Time Emotion Recognition via Attention Gated Hierarchical Memory Network. arXiv:1911.09075 [cs].
- Guimin Hu, Ting-En Lin, Yi Zhao, Guangming Lu, Yuchuan Wu, and Yongbin Li. 2022. UniMSE: Towards Unified Multimodal Sentiment Analysis and Emotion Recognition. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 7837–7851, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.
- Ashwani Bhat and Ashutosh Modi. 2022. Multi-Task Learning Framework for Extracting Emotion Cause Span and Entailment in Conversations. arXiv:2211.03742 [cs].
- Xiaohui Song, Longtao Huang, Hui Xue, and Songlin Hu. 2022. Supervised Prototypical Contrastive Learning for Emotion Recognition in Conversation. arXiv:2210.08713 [cs].
- Arijit Nandi and Fatos Xhafa. 2022. A federated learning method for real-time emotion state classification from multi-modal streaming. *Methods*, 204:340–347.

LLM based techniques

- Fu, Yumeng. "CKERC : Joint Large Language Models with Commonsense Knowledge for Emotion Recognition in Conversation." arXiv, 12 Mar. 2024, doi:10.48550/arXiv.2403.07260.
- Lei, Shanglin, et al. "InstructERC: Reforming Emotion Recognition in Conversation with a Retrieval Multi-task LLMs Framework." arXiv, 21 Sept. 2023, doi:10.48550/arXiv.2309.11911.

Pre-requisites:

- Get familiar with Conversations and Emotions data:
 - **Things you should be able to answer after understanding conversation data:**
 - Difference between different parts of a conversation in a standard dataset: dialogues, utterances, speakers, emotion-labels, cause-spans, cause-utterances and context windows.
 - *Resources*
 - Carlos Busso, Murtaza Bulut, Chi-Chun Lee, Abe Kazemzadeh, Emily Mower, Samuel Kim, Jeannette N. Chang, Sungbok Lee, and Shrikanth S. Narayanan. 2008. **IEMOCAP**: interactive emotional dyadic motion capture database. *Language Resources and Evaluation*, 42(4):335–359.
 - S. Poria, D. Hazarika, N. Majumder, G. Naik, R. Mihalcea, E. Cambria. **MELD**: A Multimodal Multi-Party Dataset for Emotion Recognition in Conversation. (2018)
 - **CMU-MOSI and CMU-MOSEI**: Zadeh, Amir, et al. "Multi-attention Recurrent Network for Human Communication Comprehension." *arXiv*, 3 Feb. 2018, doi:10.48550/arXiv.1802.00923.
 - <https://drive.google.com/drive/folders/1uEK737LXB9jAlf9kyqRs6B9N6cDncodq>
 - [mosei](#)
 - MaSaC: Bedi, Manjot, et al. "Multi-Modal Sarcasm Detection and Humor Classification in Code-Mixed Conversations." *IEEE Trans. Affective Comput.*, vol. 14, no. 02, 1 Apr. 2023, pp. 1363-75, doi:10.1109/TAFFC.2021.3083522.
 - We have this dataset locally too
- Get familiar with Speech Processing systems:
 - *Things you should be able to answer:*
 - Different parts of audio speech: low-level (pitch, tonality and tempo) and high-level (prosody, emotion interspeech)
 - Understanding different audio input formats and the difference between them, mostly on the lines of compression-ratio and audio-quality (Optional as we have dedicated audio formats for the datasets we are working on)
 - Executing speech processing pipelines to extract the above as vectors
 - Audio tokenization:
 - Splitting audio into tokens to pass them through a transformer
 - Creating a Speech recognition pipeline
 - *Resources*
 - OpenSmile:
<https://audeering.github.io/opensmile/get-started.html#default-feature-sets>
 - Radford, Alec, et al. "Robust Speech Recognition via Large-Scale Weak Supervision." *arXiv*, 6 Dec. 2022, doi:10.48550/arXiv.2212.04356.

- <https://github.com/openai/whisper>
 - <https://huggingface.co/openai/whisper-large-v3>
 - An intuitive tutorial to Mel spectrogram: ketan. "Audio Deep Learning Made Simple - Why Mel Spectrograms perform better." Ketan Doshi Blog, 19 Feb. 2021, ketanhdoshi.github.io/Audio-Mel.
- Get familiar with the Neural Machine Translation Pipeline:
 - Things you should be able to **answer** after understanding neural machine translation:
 - How many data samples are required to build a working NMT system? (in general, the more data, the better the translation; the interesting thing is to obtain the minimum threshold for an NMT system.)
 - How do we check if the dataset is good enough? (is the data sufficient for general language translation? Does the data cover a diverse range of topics? It is better to have a look at various text topics/domains general machine translation benchmarks consider [news, financial, medical, legal, etc.]
 - How to train a transformer-based system from scratch? [It is better to have the entire pipeline in mind, tokenization→training→inference, how to handle unknown tokens, etc.]
 - When to use transfer learning models?
 - *Resources:*
 - <https://joeynmt.readthedocs.io/en/latest/>
 - OpenNMT [<https://opennmt.net/>]:
 - Look at the entire pipeline, from tokenization to training from raw text pairs.
 - See if you can reproduce their results
 - **NLP-progress** [https://nlpprogress.com/english/machine_translation.html]
 - **Why You Should Do NLP Beyond English** [<https://www.ruder.io/nlp-beyond-english/>]
 - **True Zero-shot MT (an interesting read to learn from low resource)**[<https://newsletter.ruder.io/p/true-zero-shot-mt>]
 - Understanding Seq2Seq models:
 - https://lena-voita.github.io/nlp_course/seq2seq_and_attention.html
 - <https://jalammar.github.io/illustrated-transformer/>
 - <https://nlp.seas.harvard.edu/annotated-transformer/> (with code)
- Get familiar with the evaluation metrics used in emotion recognition:
 - Things you should be able to **answer** after understanding neural machine translation:
 - How to quantify the emotions of utterances?
 - Accuracy, F1, weighted-F1, Macro-F1, Micro-F1, Precision, Recall, Receiver operating Characteristics (ROC) and area-under ROC? (try to search and list them all)
 - Do you believe these metrics are able to measure the emotions in an utterance? Or, can we go more fine-grained

- Using different tokenizations or audio processing systems/pipelines can change the value of the metrics. But why?
- What's the right way of running an evaluation?
- Which metrics have stochasticity in nature?
- Which metrics are dependent on another model? Does the model's performance matter when evaluating?
- *Resources:*
 - Lever, Jake, et al. "Classification evaluation - Nature Methods." Nature, vol. 13, 28 July 2016, pp. 603-04, doi:10.1038/nmeth.3945.
 - M, Hossin, and Sulaiman M.N. "A Review on Evaluation Metrics for Data Classification Evaluations." International Journal of Data Mining & Knowledge Management Process, vol. 5, no. 2, Mar. 2015, pp. 01–11. DOI.org (Crossref), <https://doi.org/10.5121/ijdkp.2015.5201>.

Target a replicable baseline:

Joshi, Abhinav, et al. "COGMEN: COntextualized GNN based Multimodal Emotion recognitionN." ACL Anthology, July 2022, pp. 4148-64, doi:10.18653/v1/2022.naacl-main.306.

- Replicate the code for the given datasets
- Extend it to the remaining datasets (IEMOCAP, MELD, MOSEI, MaSaC [hindi-english code-mixed])
- Convert it to a real-time setting by removing future utterances in the context window of each utterance node in the graph (You'll understand this once you go through the paper)

Guimin Hu, Ting-En Lin, Yi Zhao, Guangming Lu, Yuchuan Wu, and Yongbin Li. 2022. UniMSE: Towards Unified Multimodal Sentiment Analysis and Emotion Recognition. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 7837–7851, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.

- Similar steps as above

Rough four-week plan for the project.

Week 1 (getting things ready):

- Replicate the baseline results. [See if there are any missing gaps in results replication, analyze where the model fails, and determine whether there is a distribution shift between the provided train validation and test sets. In short, prepare the pipeline and understand the codebase well for more straightforward modifications in the future.]
 - Prepare the pipeline to train any dataset when we change the model.
- Prepare a speech-to-text-to-emotion pipeline [dataset → evaluation scores]
 - We may receive data with audio and transcribed text
- Use any other ASR system to do the above step
 - Just so that we can prove that real-time emotion recognition can be done with audio only
- Pipeline on speaker diarization

Milestone

- We are able to run baselines in the below real-time modes:
 - Audio only
 - Audio + transcribed/annotated text
 - Audio + ASR text
 - Transcribed text only
 - AST text only

